# Master Module Virtual Product Development

**Project Phase 2**

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# *OUTLINE*

1. **Design Adaptation** – Incorporate additional requirements.

2. **Topology Optimization** – Lightweight, stiff structure.

3. **CAD Detailing** – Full assembly, constraints, parametrics.

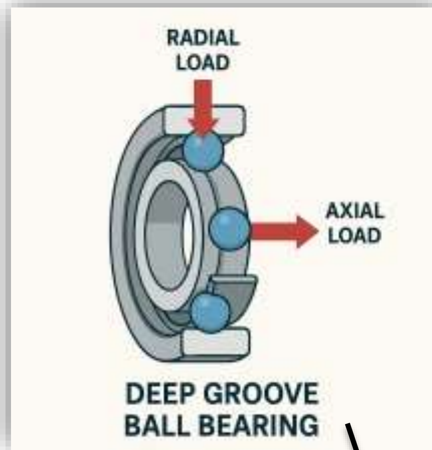
4. **Stress & Deformation Analysis.**

5. **Transient Printing Simulation** – sample models.

# 1. Design Adaptation – Incorporate additional requirements.

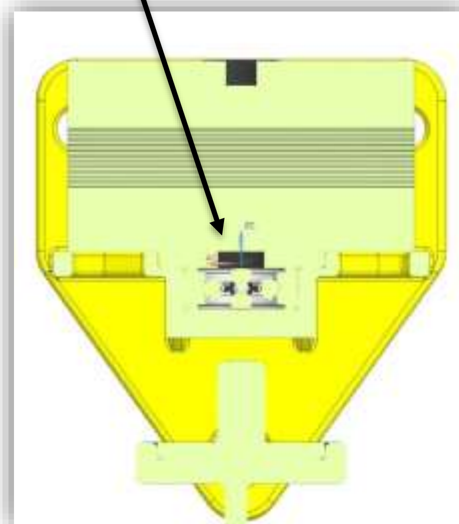
## Why Add a Bearing?

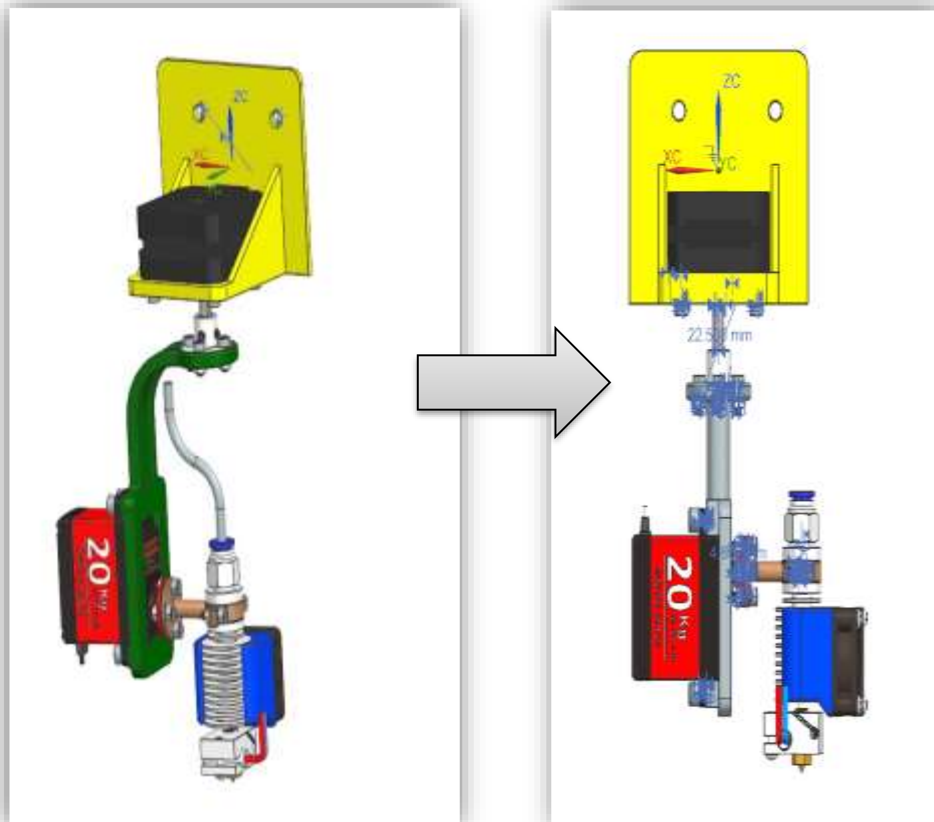
1. Reduces load on motor
2. Absorbs radial & axial forces
3. Minimizes friction for smoother motion




## Why DN 625?

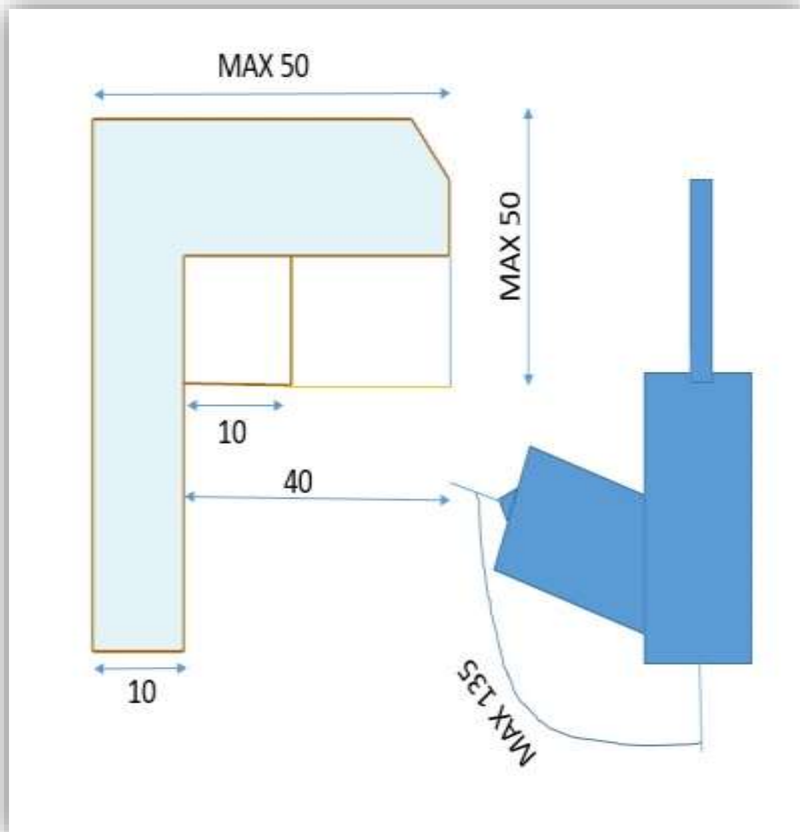
1. Handles both radial and axial loads
2. Compact and reliable
3. Ensures smooth force transmission





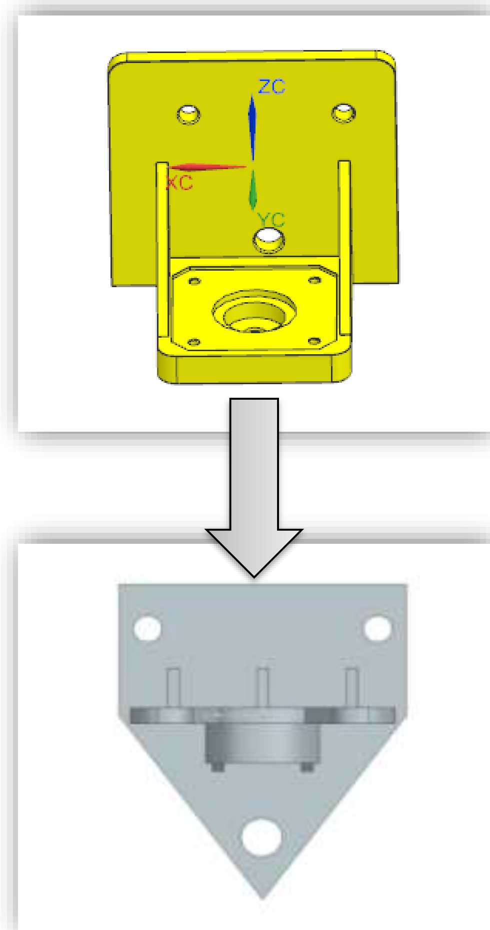
We improved the frame's stiffness to eliminate minor flex seen in tests. The new design offers greater stability and is better suited for precise, reliable performance.

- MAXIMUM PARAMETERS FOR UNDERCUTS AND OVERHANGS



| Aspect Improved | Brief Reason for Change |
|---|---|
| Frame Geometry & Material | Previous flex under load—new design improves rigidity. |
| Structural Stiffness | Better accuracy and less vibration. |
| Load Distribution | More balanced stress across joints. |
| Long-Term Reliability | Stronger frame reduces fatigue and maintenance. |
| Performance Under Stress | Improved repeatability and dynamic response. |

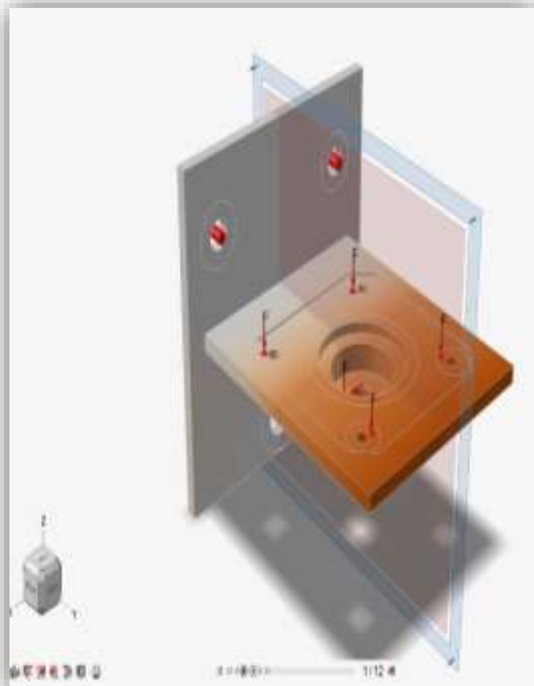
We replaced the original support plate to reduce overall height, enabling a taller build volume. This adjustment expanded the 3D printer's usable area without compromising structural integrity.



# 2. Topology Optimization– Lightweight, stiff structure.

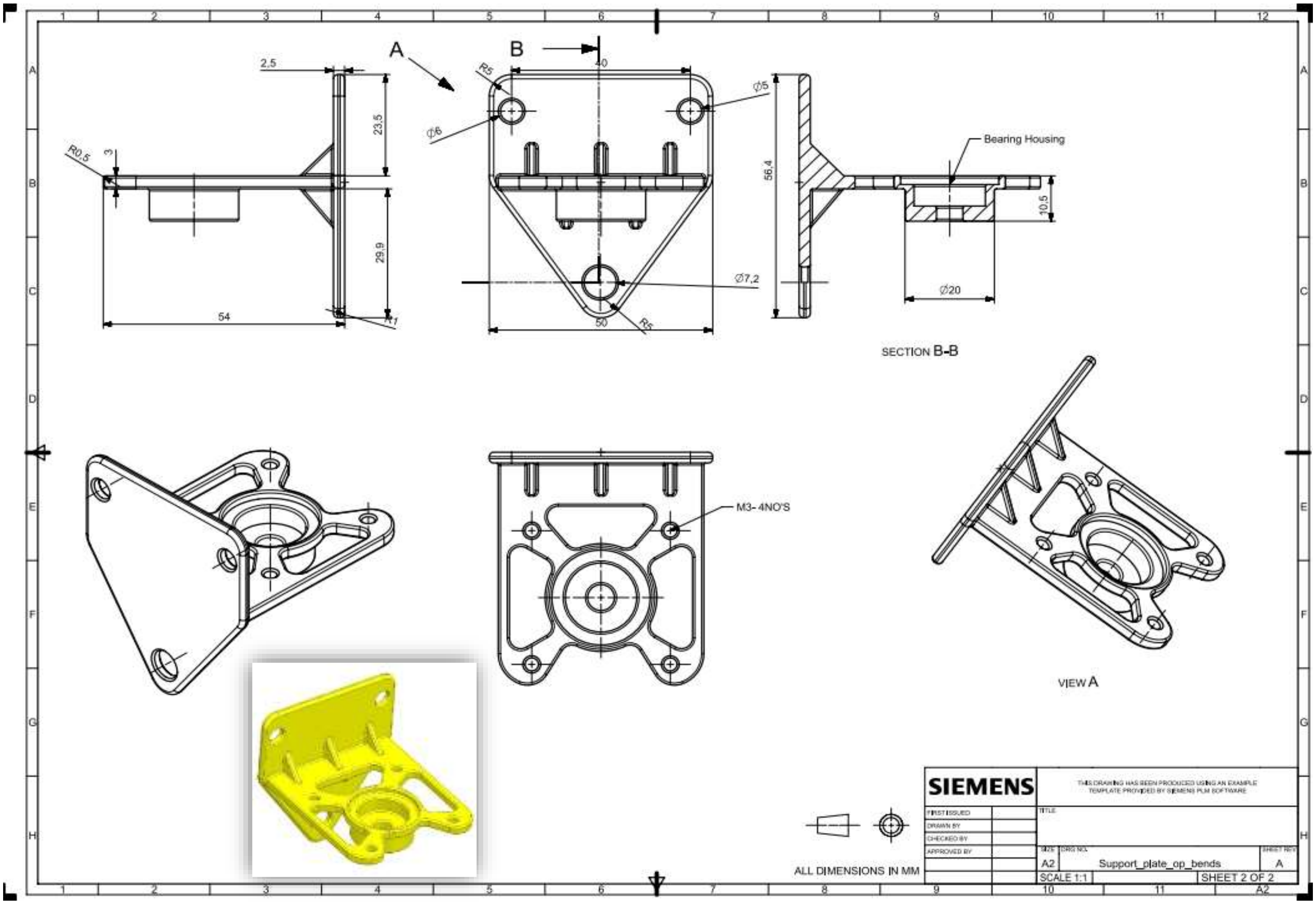
## 1. SUPPORT PLATE

Old Part



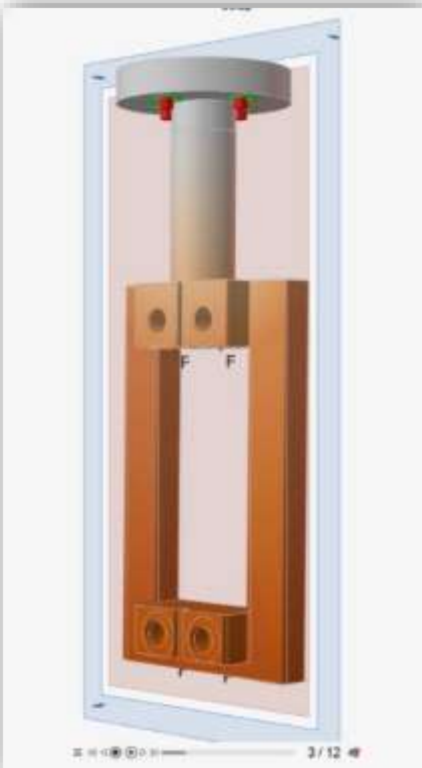
| Description | Before | After |
|---|---|---|
| Mass | 0.0463 Kg | 0.0282 Kg |
| Volume | 17061.88 mm^3 | 10416.12 mm^3 |
| Material | Aluminum (6061-T6) | |

Optimized Part


A
B
2,5
R5
40
Ø5
Ø6
23,5
R0,5
3
Bearing Housing
56,4
10,5
29,9
Ø7,2
Ø20
54
R1
50
R5
SECTION B-B
M3- 4NO'S
VIEW A
ALL DIMENSIONS IN MM
SIEMENS
THIS DRAWING HAS BEEN PRODUCED USING AN EXAMPLE TEMPLATE PROVIDED BY SIEMENS PLM SOFTWARE
FIRST ISSUED
DRAWN BY
CHECKED BY
APPROVED BY
TITLE
SIZE
DRG NO.
A2
Support_plate_op_bends
A
SCALE 1:1
SHEET 2 OF 2

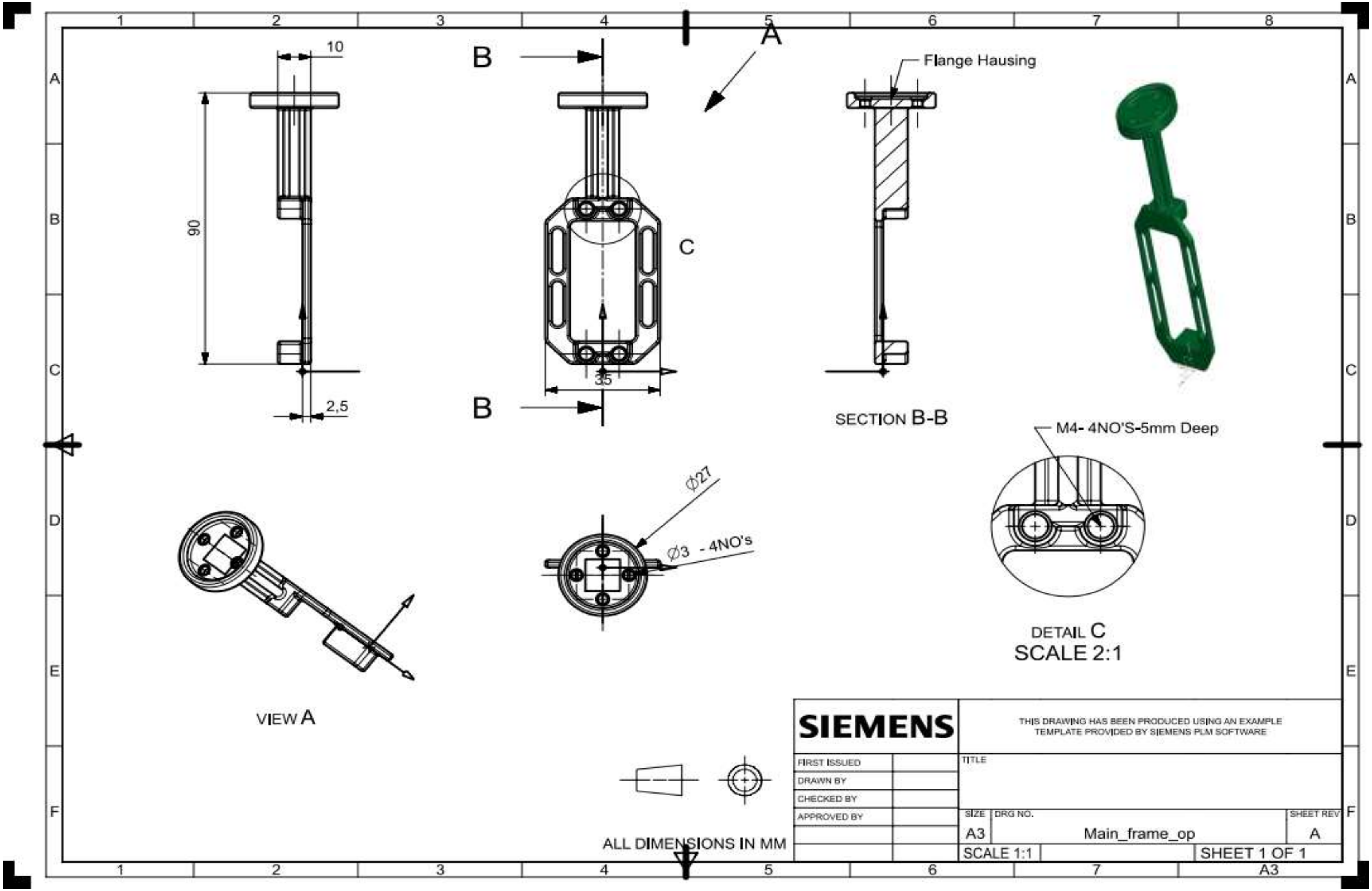
## 2. MAIN FRAME

Old Part



| Description | Before | After |
|---|---|---|
| Mass | 0.0312 Kg | 0.0175 Kg |
| Volume | 11506.66 mm^3 | 6466.31 mm^3 |
| Material | Aluminum (6061-T6) | |

Optimized Part


10
90
2,5
B
A
35
C
B
Flange Hausing
SECTION B-B
M4- 4NO'S-5mm Deep
Ø27
Ø3 - 4NO's
VIEW A
DETAIL C
SCALE 2:1
ALL DIMENSIONS IN MM
SIEMENS
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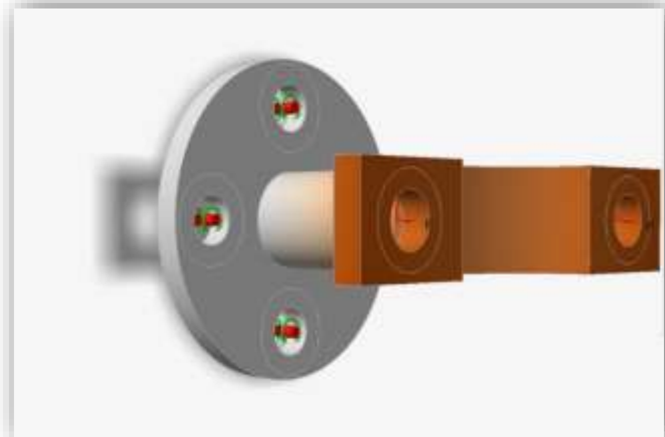
FIRST ISSUED
DRAWN BY
CHECKED BY
APPROVED BY
TITLE
SIZE
DRG NO.
SHEET REV
A3
Main_frame_op
A
SCALE 1:1
SHEET 1 OF 1
A3

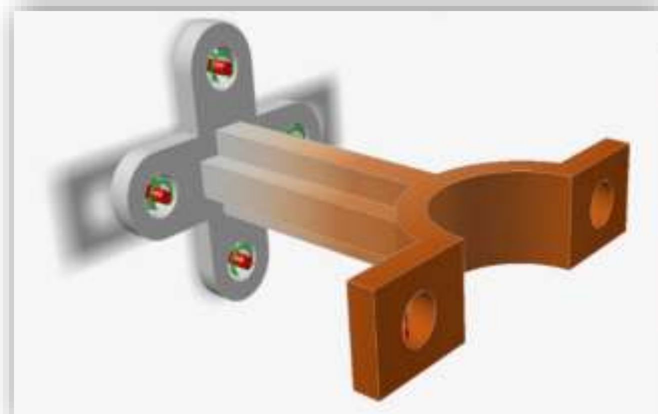
# 3. CLAMP

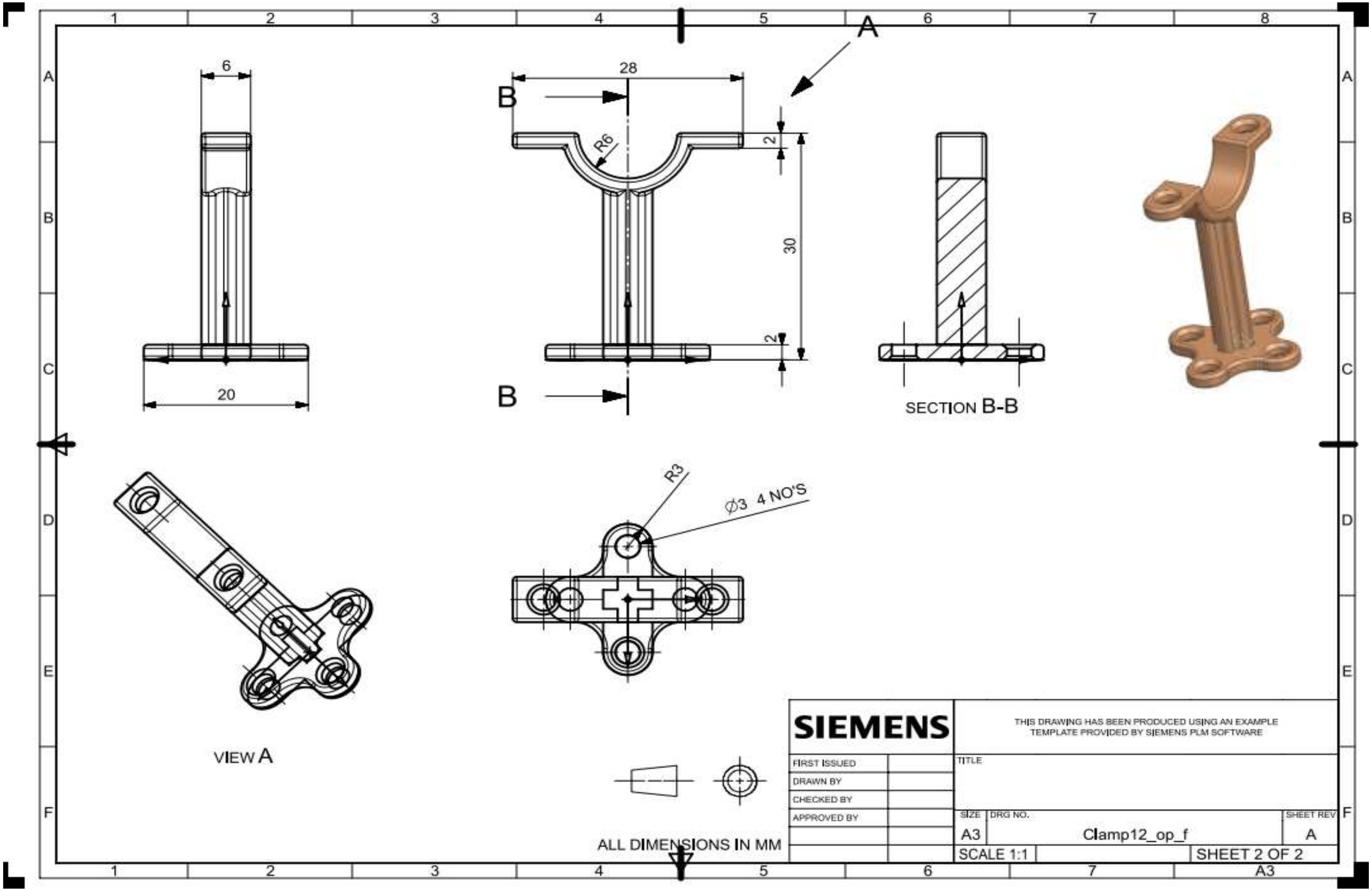
| Description | Before | After |
|---|---|---|
| Mass | 0.0041 Kg | 0.0031 Kg |
| Volume | 1511.79 mm^3 | 1140.17 mm^3 |
| Material | Aluminum (6061-T6) | |

Old Part



Optimized Part




6
28
B
A
R6
2
30
2
20
B
SECTION B-B
R3
Ø3 4 NO'S
VIEW A
ALL DIMENSIONS IN MM
SIEMENS
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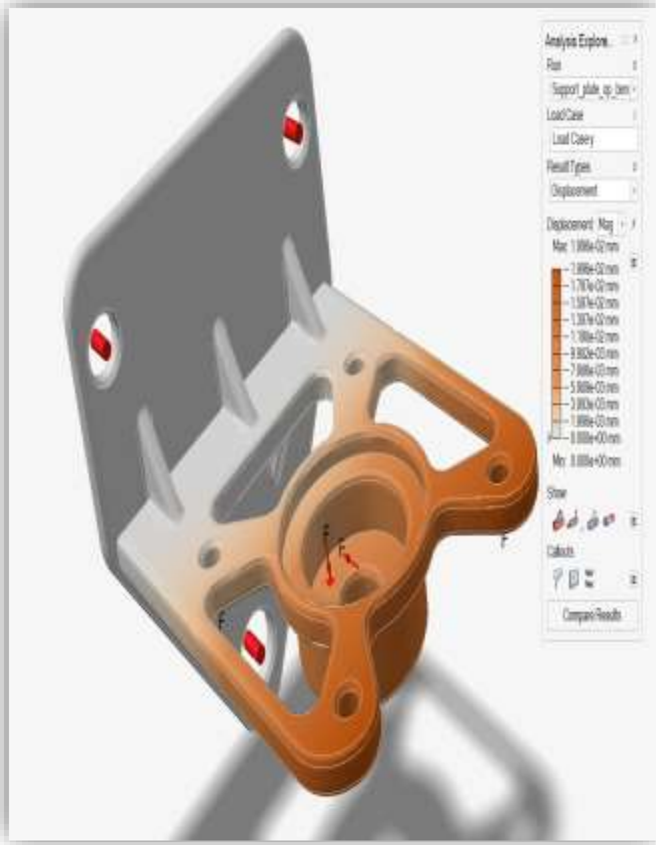
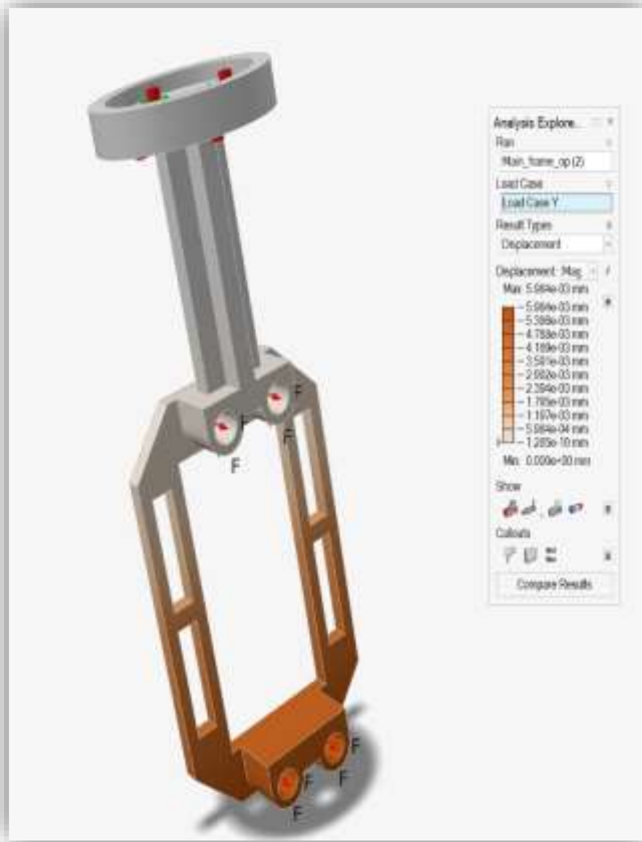
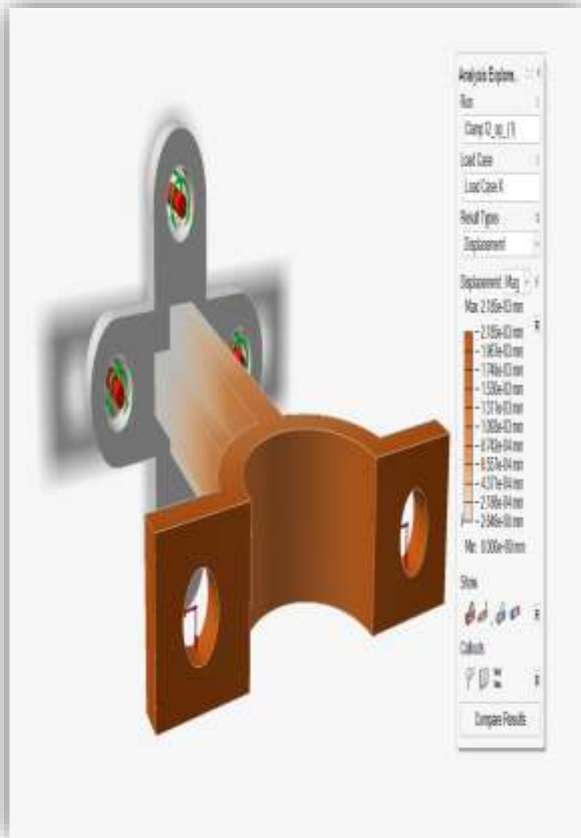
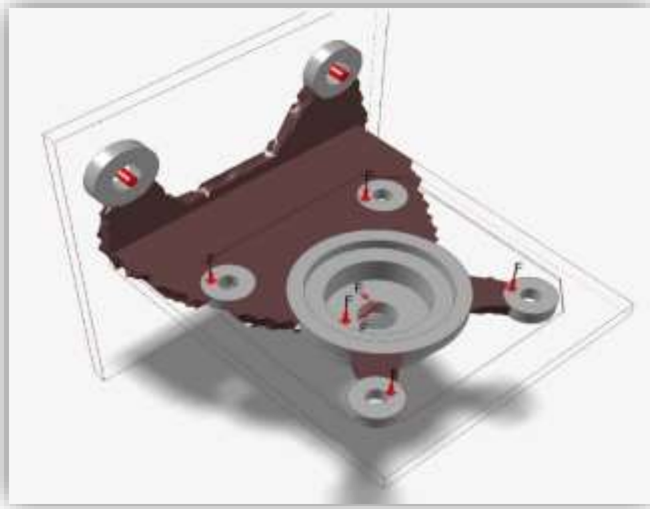
CHECKED BY
APPROVED BY
TITLE
SIZE
DRG NO.
SHEET REV
A3
Clamp12_op_f
A
SCALE 1:1
SHEET 2 OF 2

# OPTIMIZATION : DISPLACEMENT

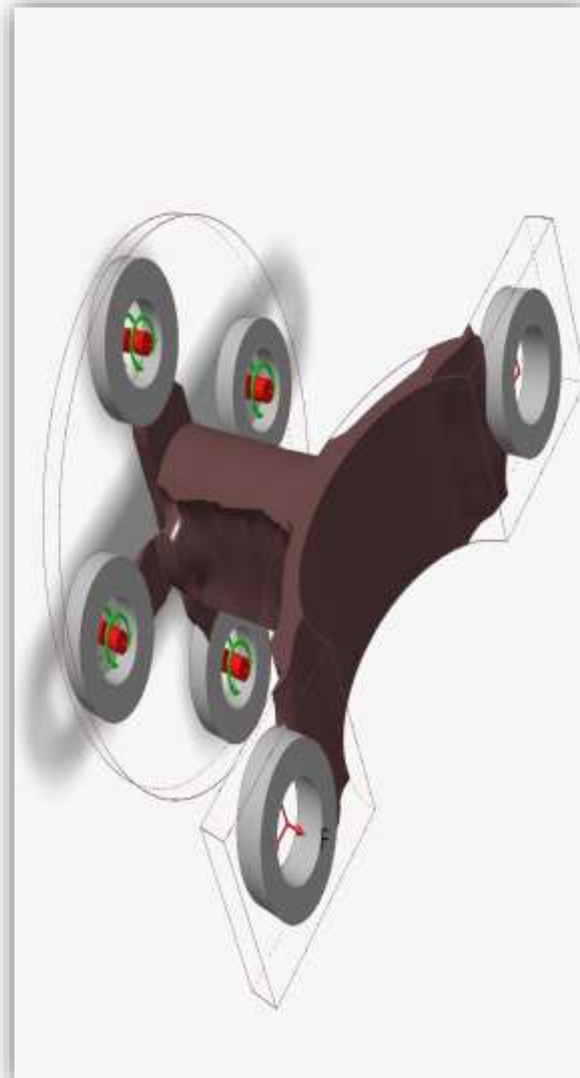



F
F
F
F
F

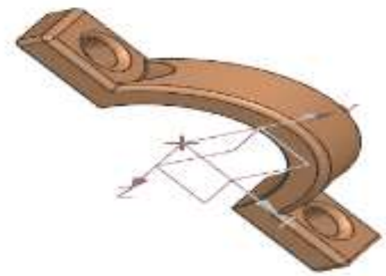

F
F
F
F


F

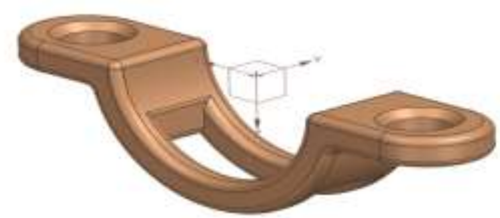
# 4. SMALL CLAMP

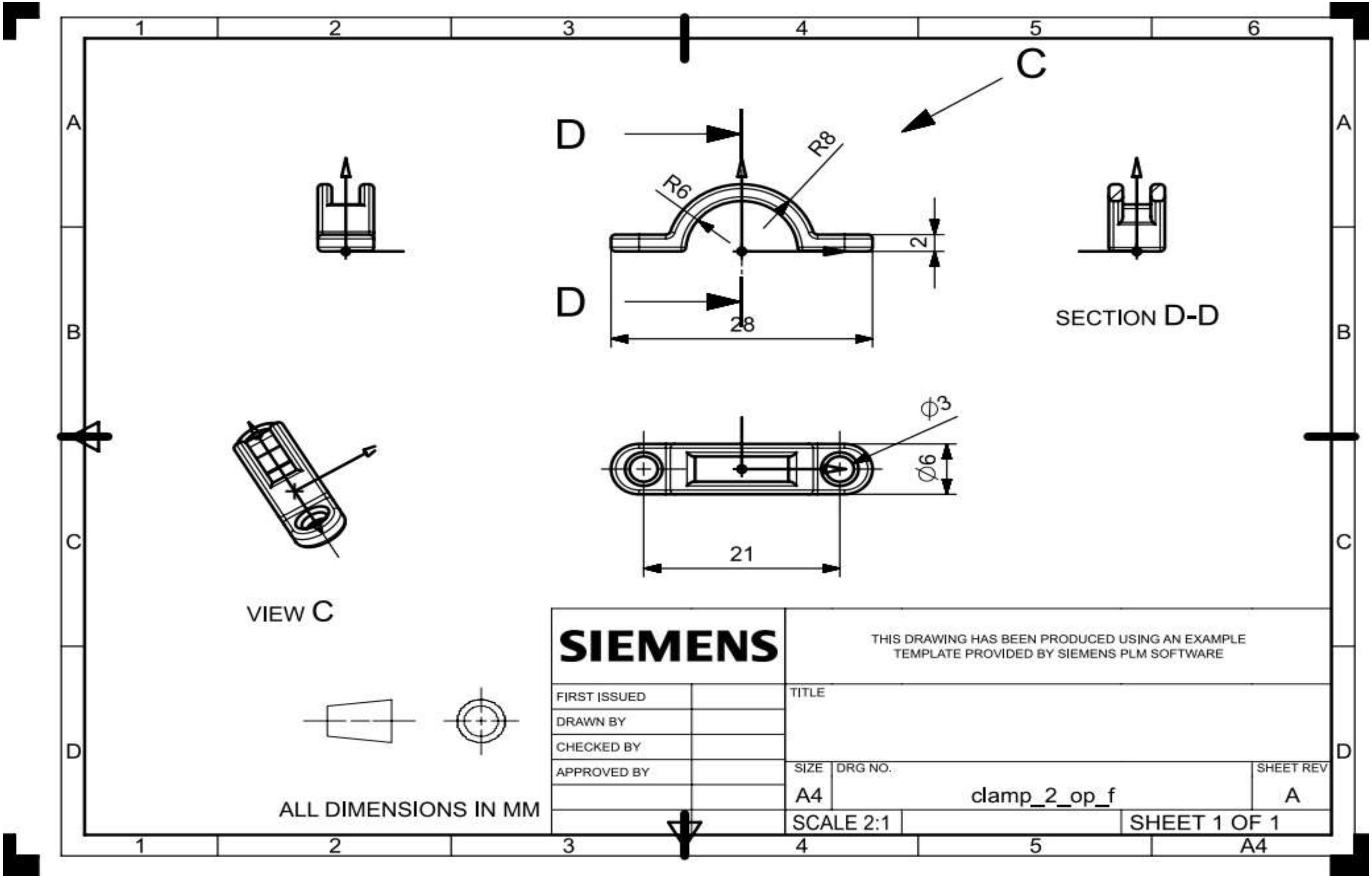
| Description | Before | After |
|---|---|---|
| Mass | 0.0010 Kg | 0.00080 Kg |
| Volume | 370.40 mm^3 | 281.2 mm^3 |
| Material | Aluminum (6061-T6) | |

Old Part



Optimized Part




C
D
D
R6
R8
2
28
SECTION D-D
ϕ3
Ø6
21
VIEW C
ALL DIMENSIONS IN MM
SIEMENS
THIS DRAWING HAS BEEN PRODUCED USING AN EXAMPLE
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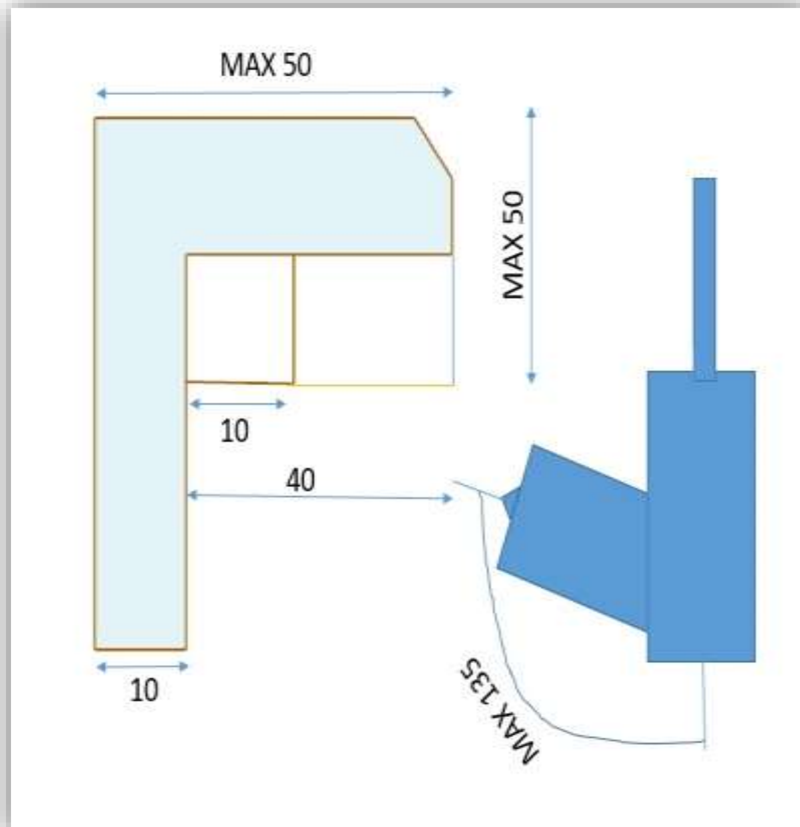
APPROVED BY
TITLE
SIZE
DRG NO.
SHEET REV
A4
clamp_2_op_f
A
SCALE 2:1
SHEET 1 OF 1
A4

# 3. CAD Detailing – Full assembly, constraints, parametrics.

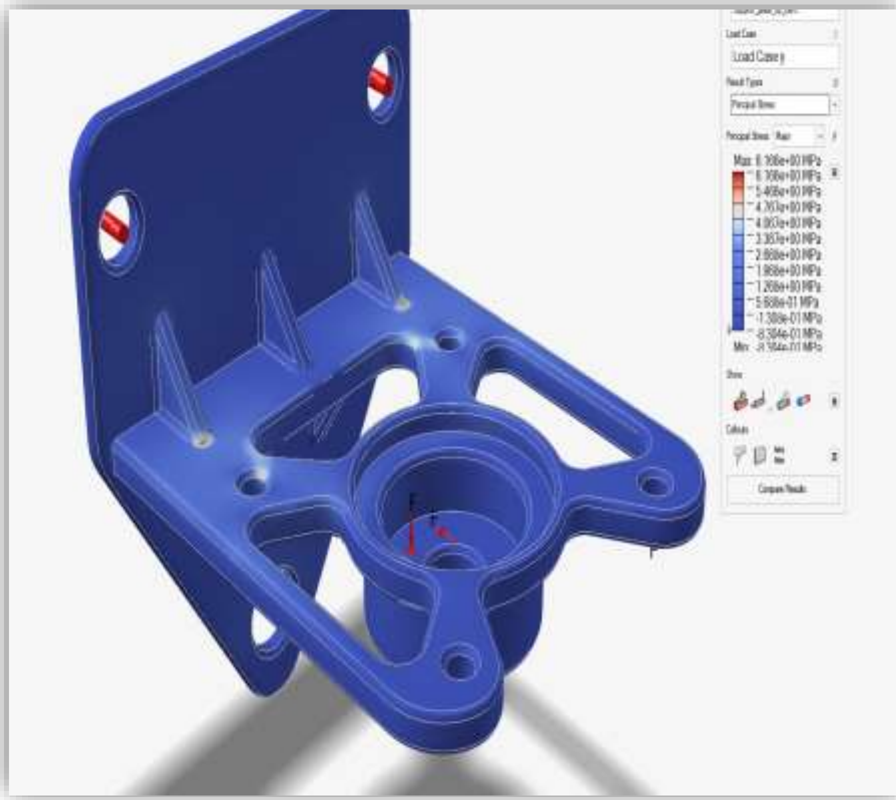
| PARAMETER | BEFORE | AFTER |
|---|---|---|
| MASS | 0.418 Kg | 0.3657 Kg |
| DIMENSIONS | 111.5*35.98*227.8 mm | 100.25*52.35*177 mm |

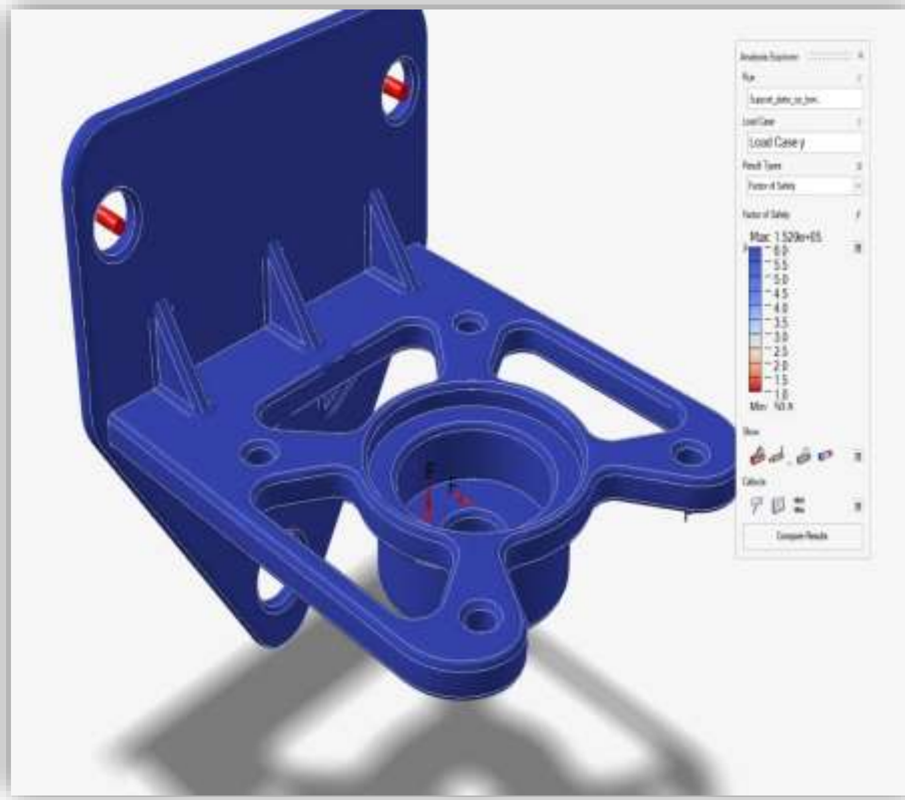
- Centre of Mass and Printing Angle

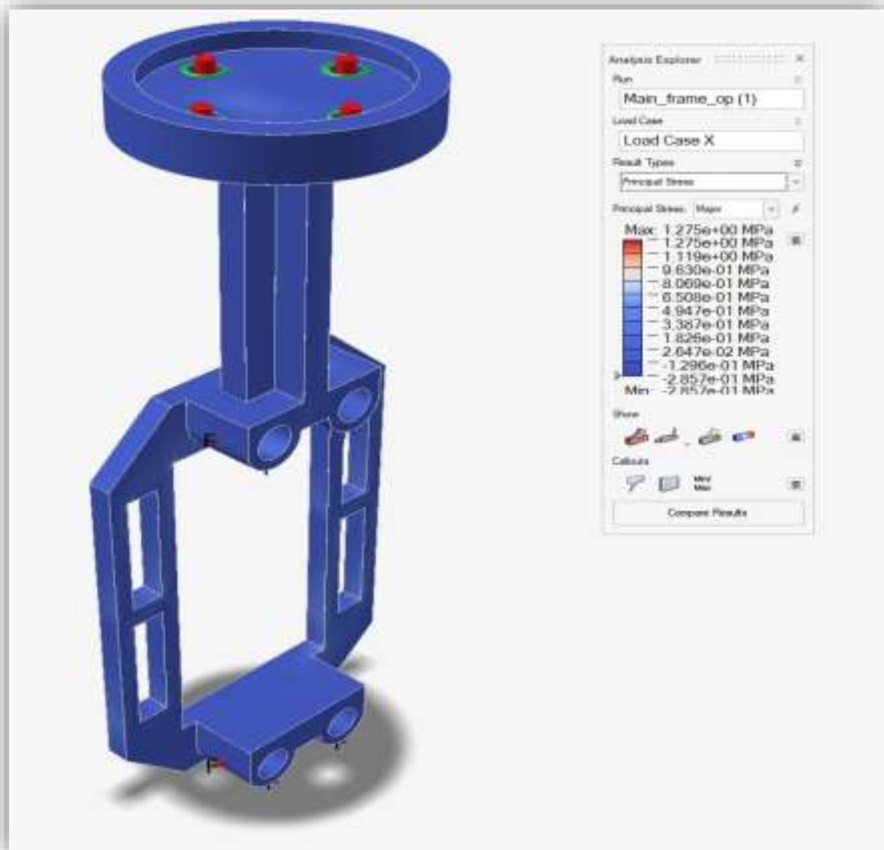
- PRINCIPAL STRESSES AND FACTOR OF SAFETY

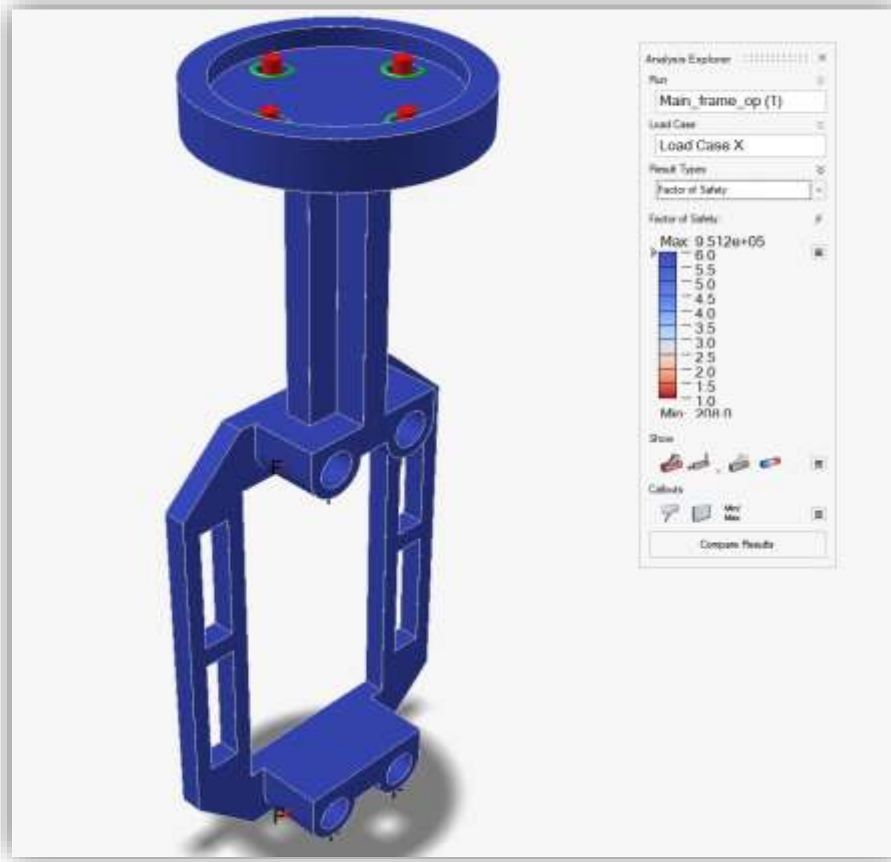
## 1. SUPPORT PLATE



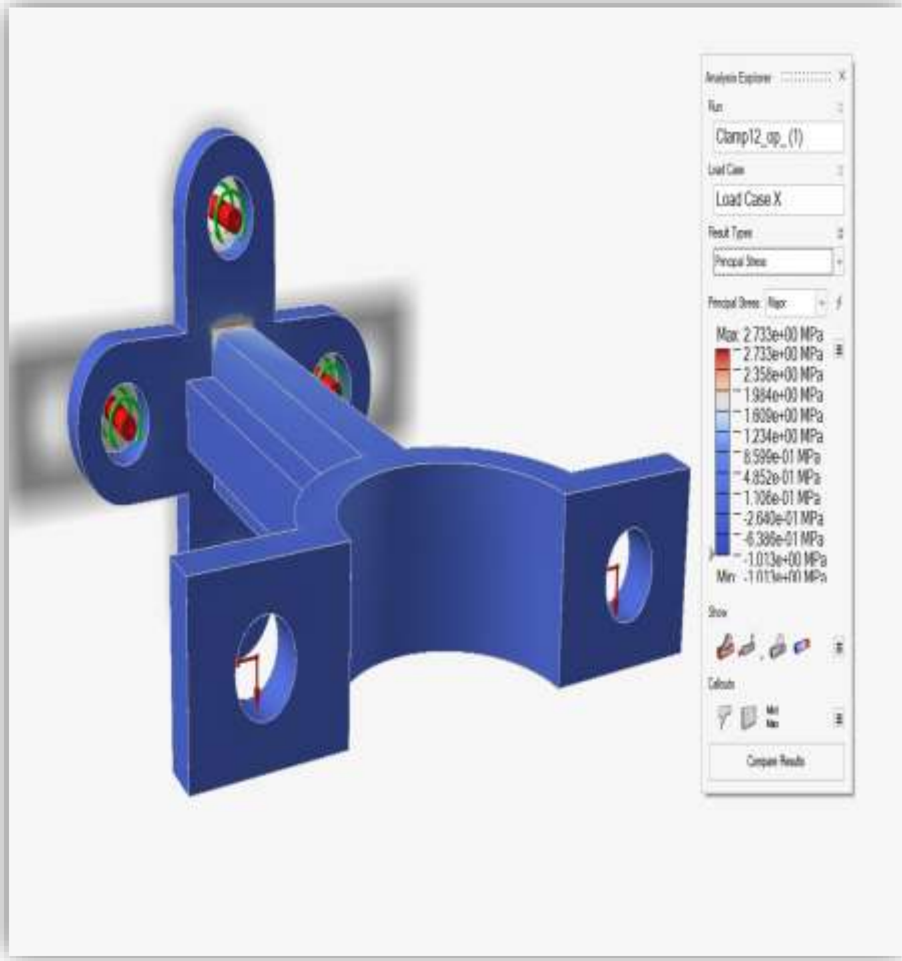



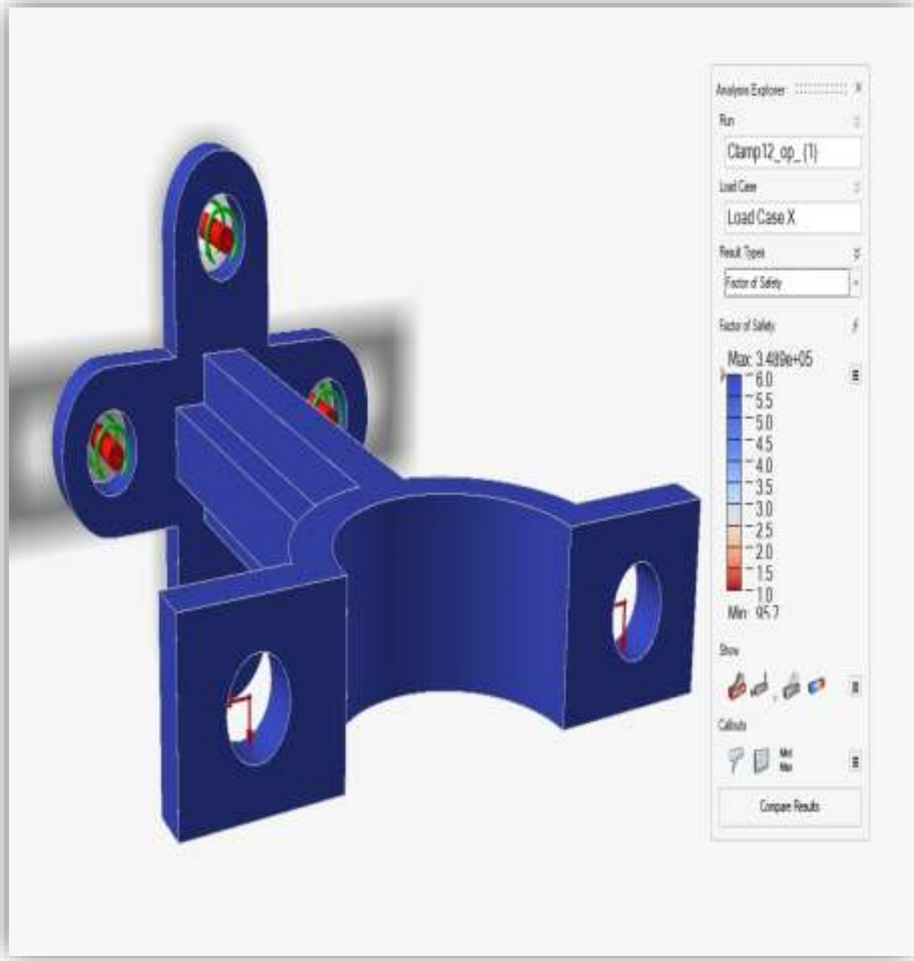
# 2. MAIN FRAME


Analysis Explorer
Run
Main_frame_op (1)
Load Case
Load Case X
Result Types
Principal Stress
Principal Stress: Major
Max: 1.275e+00 MPa
1.275e+00 MPa
1.119e+00 MPa
9.630e-01 MPa
8.069e-01 MPa
6.508e-01 MPa
4.947e-01 MPa
3.387e-01 MPa
1.826e-01 MPa
2.647e-02 MPa
-1.296e-01 MPa
-2.857e-01 MPa
Show
Callouts
Compare Results


Analysis Explorer
Run
Main_frame_op (1)
Load Case
Load Case X
Result Types
Factor of Safety
Factor of Safety
Max: 9.512e+05
6.0
5.5
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
Show
Callouts
Compare Results

# 3. CLAMP

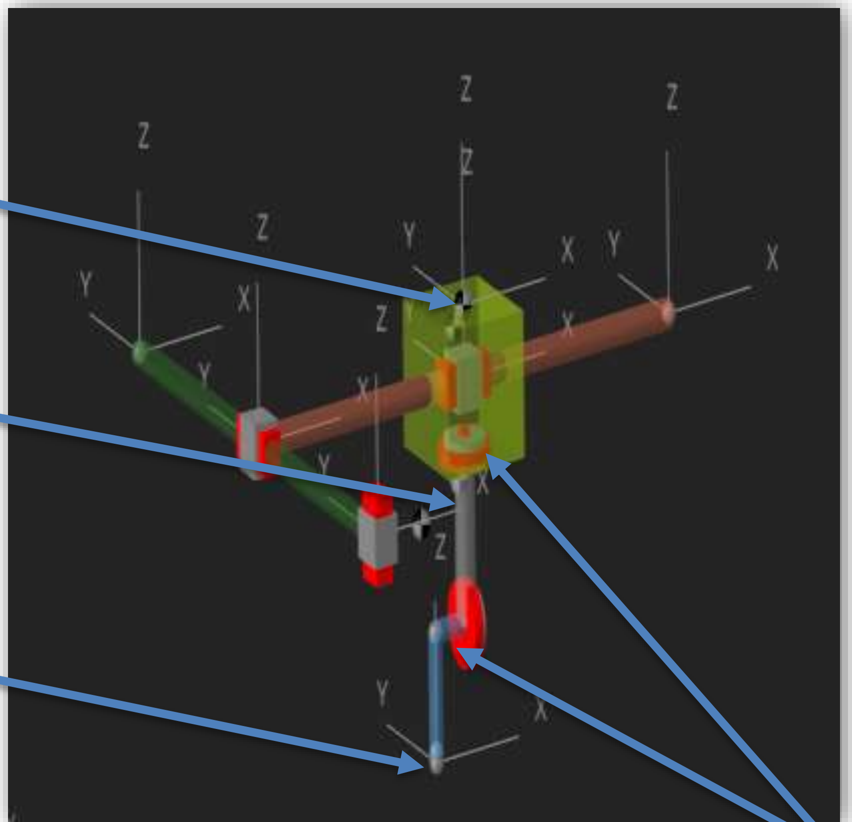

Analysis Explorer
Run
Clamp12_op_(1)
Load Case
Load Case X
Result Types
Principal Stress
Max: 2.733e+00 MPa
2.733e+00 MPa
2.358e+00 MPa
1.984e+00 MPa
1.609e+00 MPa
1.234e+00 MPa
8.590e-01 MPa
4.852e-01 MPa
1.106e-01 MPa
-2.640e-01 MPa
-6.386e-01 MPa
-1.013e+00 MPa
Min: -1.013e+00 MPa
Show
Callouts
Compare Results


Analysis Explorer
Run
Clamp12_op_(1)
Load Case
Load Case X
Result Types
Factor of Safety
Factor of Safety
Max: 3.489e+05
6.0
5.5
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
Min: 95.7
Show
Callouts
Compare Results

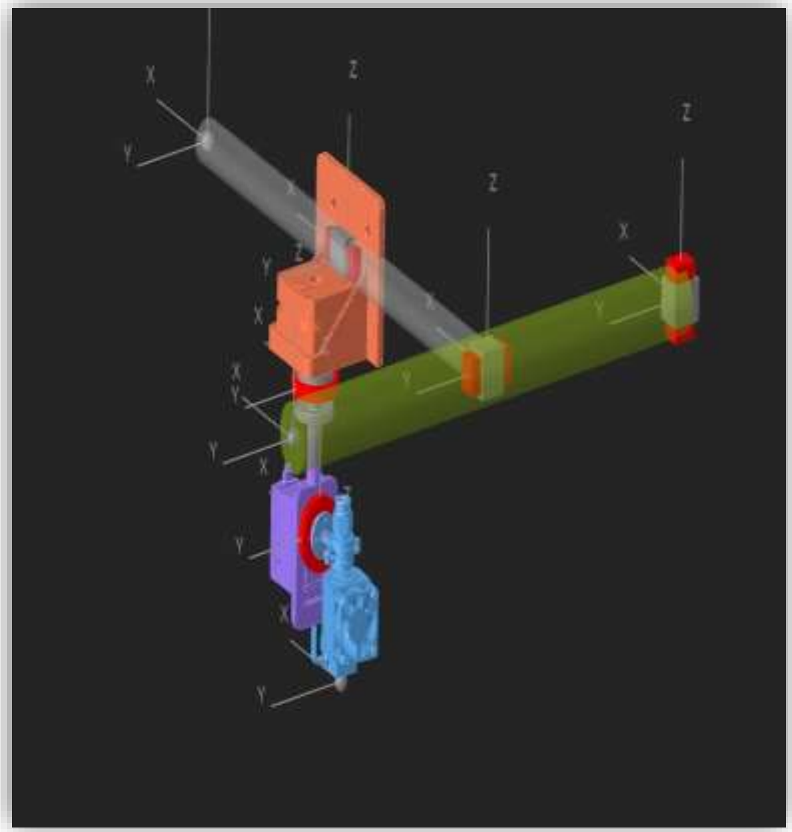
# 5. Transient Printing Simulation – sample models



Primitive Model



Imported CAD Model



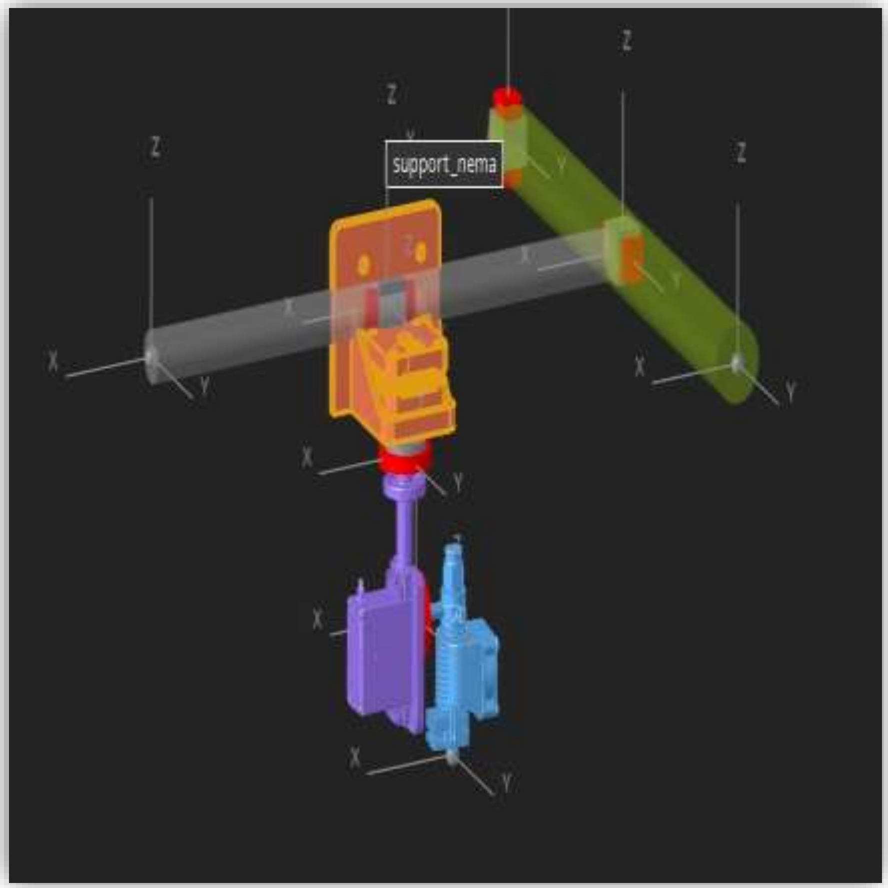
(0,0,0)

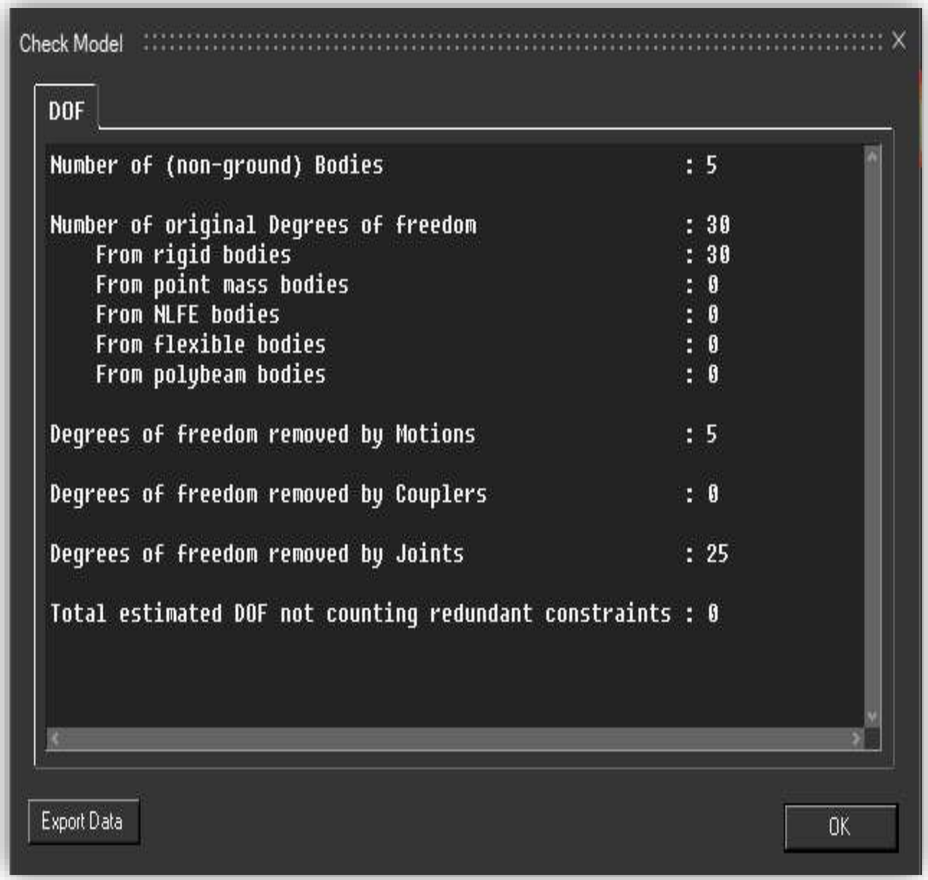
Main Frame

Marker at tip of nozzle

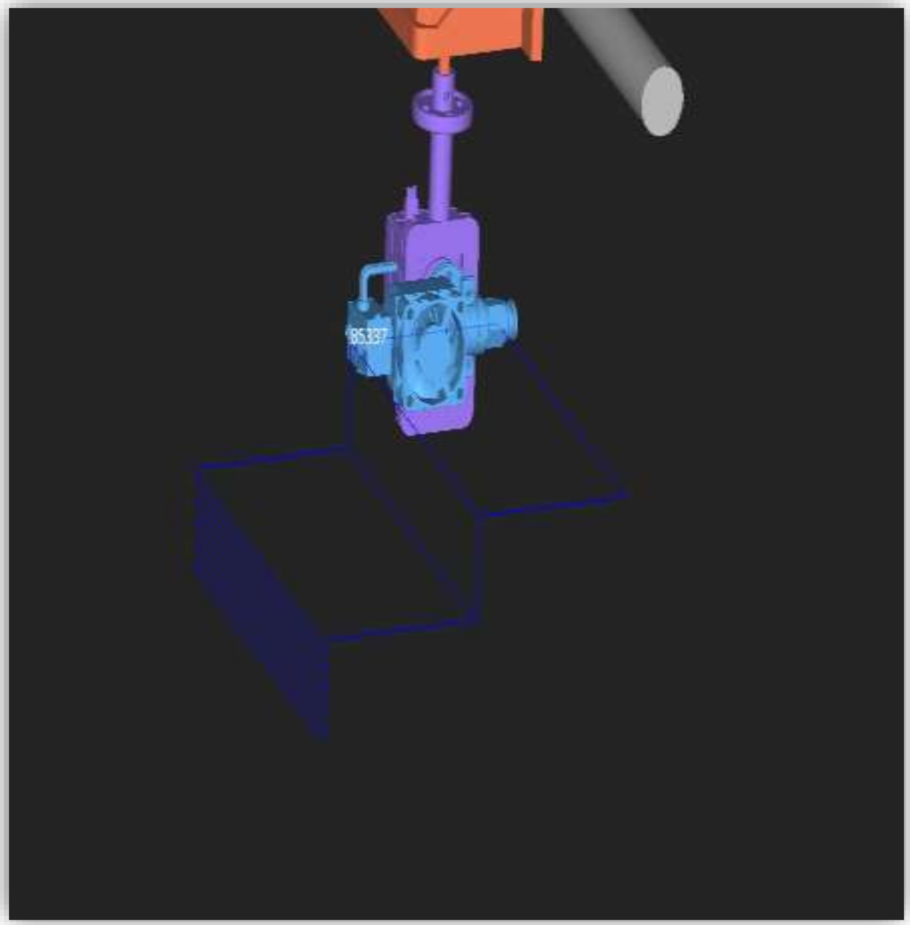
Revolute Joints

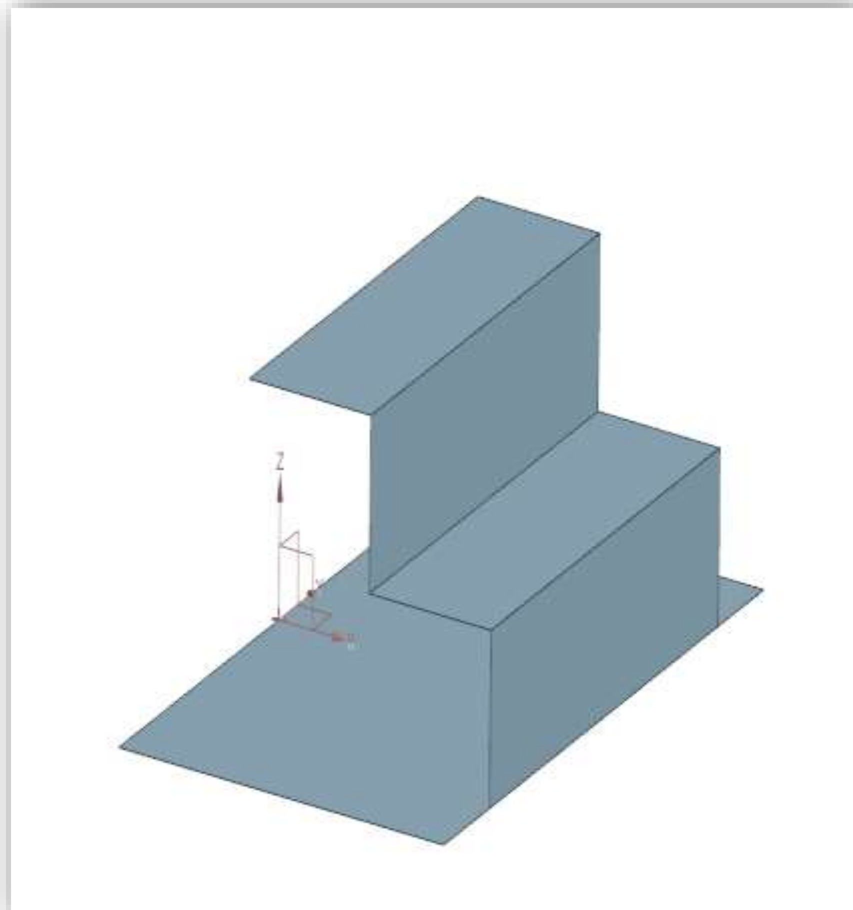
- DEGREE OF FREEDOM



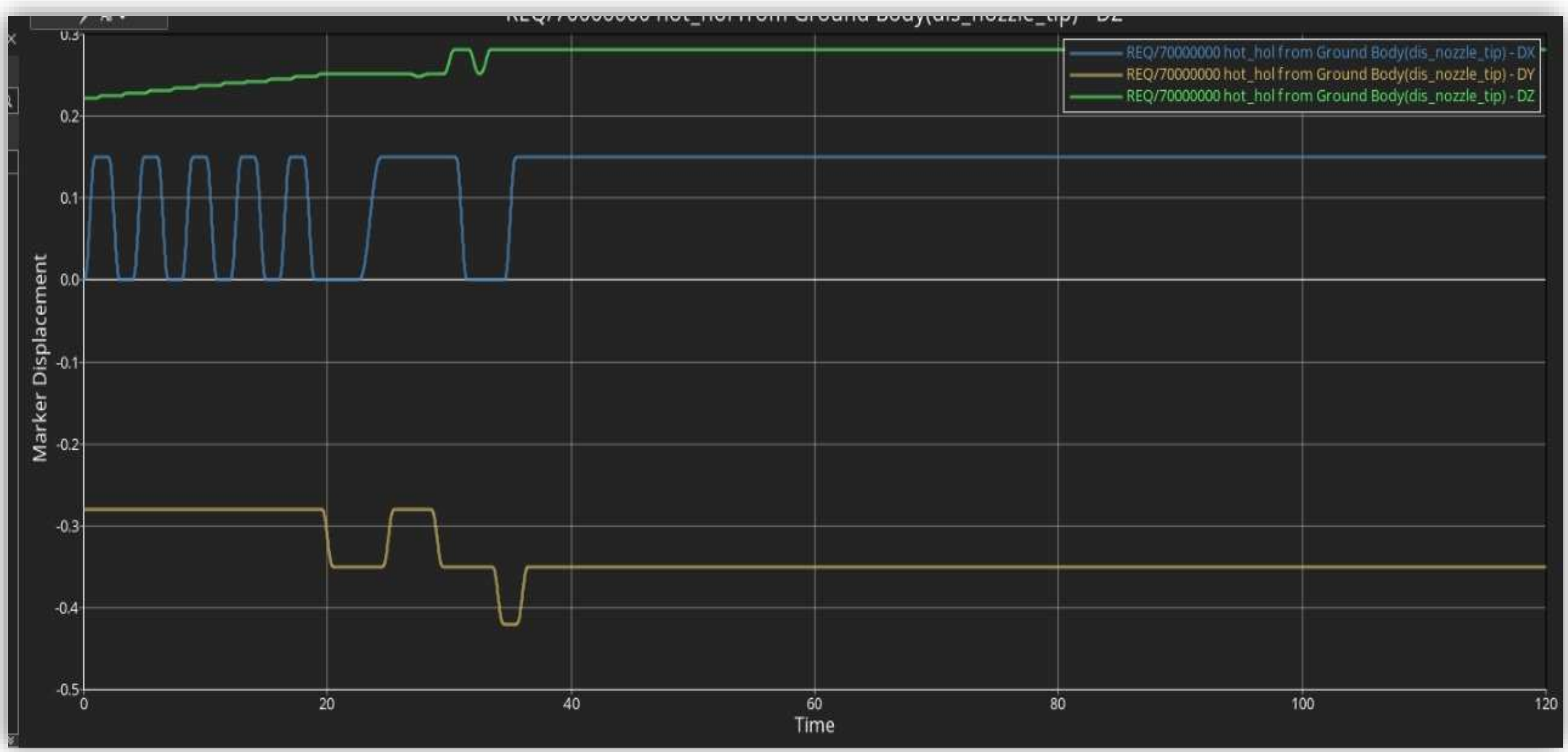



- STAIR SAMPLE

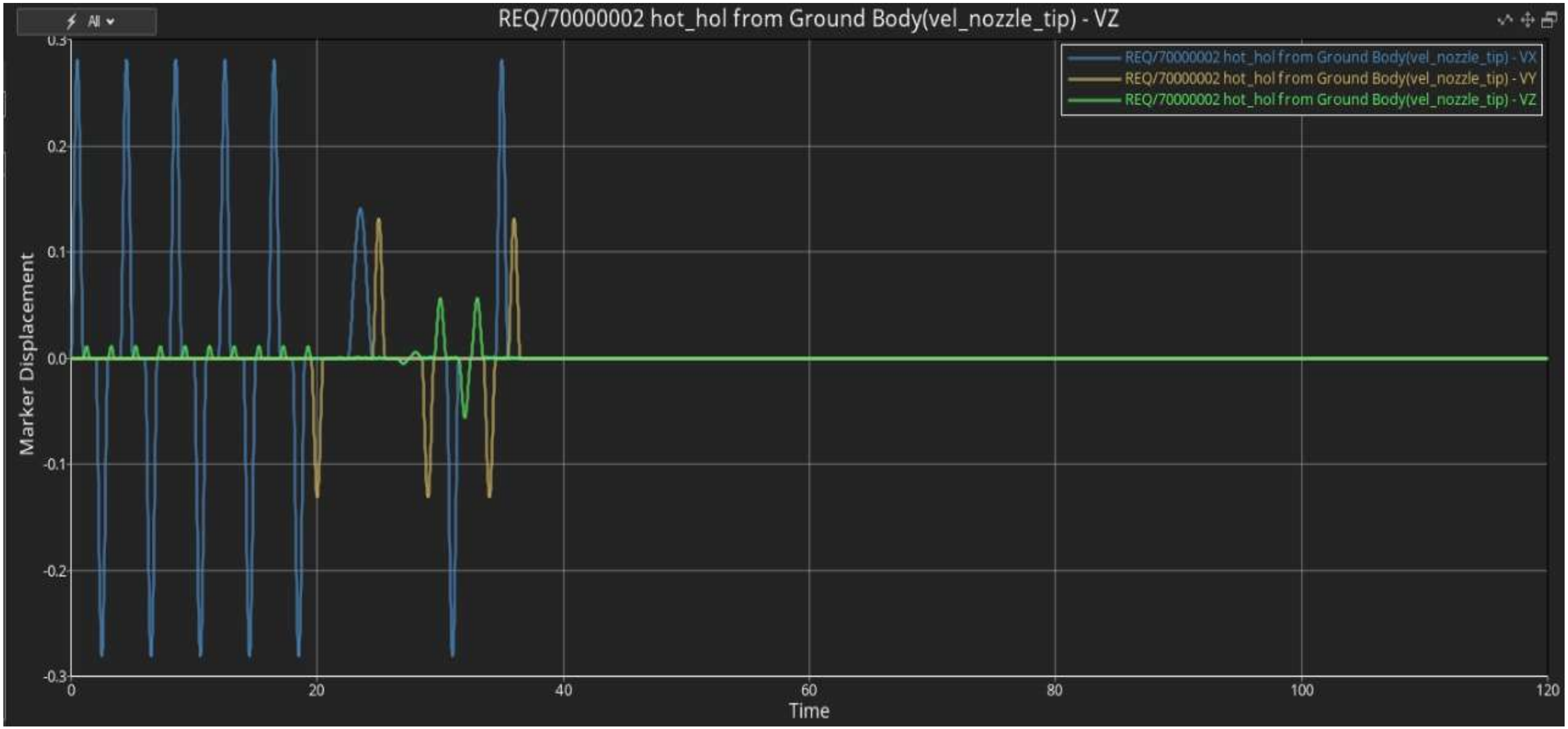


- MARKER DISPLACEMENT IN DX,DY AND DZ



- OUTPUT USING VELOCITY CURVES



THANK YOU ANY QUESTIONS!!